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To cite this article: Yurong Li *et al* 2025 *J. Phys.: Conf. Ser.* **3110** 012005

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# Design of a dual tipping-bucket rain gauge acquisition system with multi-source sensor fusion

Yurong Li<sup>1</sup>, Haoyu Wen<sup>2</sup>, Xu Chen<sup>1</sup>, Yuexiang Wang<sup>1</sup>, Bin Wen<sup>1\*</sup>

<sup>1</sup>Chengdu University of Information Technology, Sichuan, Chengdu, 610225, China

<sup>2</sup>Southwest Jiaotong University, Sichuan, Chengdu, 611756, China

\*Corresponding author's e-mail: 1158064191@qq.com

**Abstract.** Driven by climate change, rainstorms have elevated the importance of precipitation measurement accuracy. Dual tipping-bucket rain gauges represent an enhanced version of traditional single-bucket designs, engineered to deliver higher-precision rainfall monitoring. However, empirical evidence reveals that their stability and accuracy consistently fall below industry requirements. To address these limitations, this paper proposes a multi-source sensor fusion solution for dual tipping-bucket rain gauges. The system employs an STM32 microcontroller as the central processor and integrates a multi-network convergence routing module for data transmission. Through synergistic incorporation of infrared sensors, photoelectric rain sensors, and USB cameras, thereby significantly enhancing precipitation measurement precision.

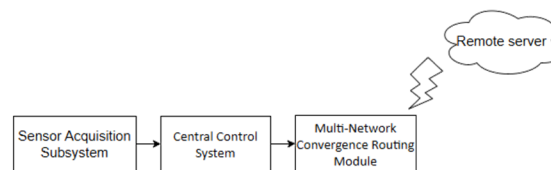
## 1. Introduction

Due to global climate change, rainfall is changing [1], placing higher demands on precise, real-time meteorological monitoring. Flood disasters begin with high rainfall [2]. Rainfall monitoring and data are of interest for many human activities, such as irrigation management, hydrological monitoring, water resources management, flood forecasting, and flood alert systems [3]. Tipping-bucket rain gauges, as instruments for measuring precipitation, require robust durability and are consequently widely deployed in various automatic weather stations and environmental monitoring systems [4]. Currently, as the primary equipment for rainfall measurement, dual tipping-bucket rain gauges are better suited for high-intensity precipitation [5]. However, their stability and accuracy still fall significantly short of industry requirements. Particularly during thunderstorms, heavy rainfall, or strong electromagnetic interference, these systems are prone to signal interference and false triggering, mechanical vibrations or shocks, and damage to electronic components, resulting in substantial data inaccuracies. To address these inherent limitations of traditional tipping-bucket rain gauges, rectify rainfall data errors, meet the demands of modern technological advancements, and enhance the intelligence and operational efficiency of rainfall monitoring systems [6], this paper applies embedded technology to develop an adjacent installed rainfall acquisition system for dual tipping-bucket rain gauges.

## 2. System Framework Structure

As shown in Figure 1, the acquisition subsystem collects sensor data and transmits it to the main control system via serial communication. The main control system performs fusion analysis on the data and forwards required information to remote servers through the multi-network converged routing module for further processing.





**Figure 1.** Architecture of Multi-source Sensor Fusion-based Double-bucket Rain Gauge Acquisition System

### 3. System Hardware Circuit Design

#### 3.1. Sensor Acquisition Circuit Module Design

The sensor acquisition circuit module is equipped with an STM32F103C8T6 chip as the MCU, primarily responsible for collecting and preliminary processing signals from two pairs of infrared emitter-receiver pairs and pulse sensors, ultimately transmitting data to the central controller. Its circuit must include essential subsystems: power supply circuit, clock circuit, reset circuit, and programming circuit [4]. Sensor signals are acquired using the MCU's built-in 12-bit ADC. When the tipping bucket obstructs the infrared beam between paired emitters and receivers, the receiver outputs a high-level signal [7]. The physical implementation is shown in Figure 2.



**Figure 2.** Physical Implementation of Sensor Subsystem Circuit.

#### 3.2. Design of Multi-Network Convergence Routing Module

The multi-network convergence module adopts a hardware combination of Quectel's 4G communication module EC200U and STM32F4 microcontroller. The EC200U, a 4G module from Quectel, features built-in multiple protocols that reduce development costs [8]. By connecting to a multi-network SIM card, it enables access to diverse networks and functions as a USB network adapter, interfacing with the STM32. The physical implementation is shown in Figure 3.

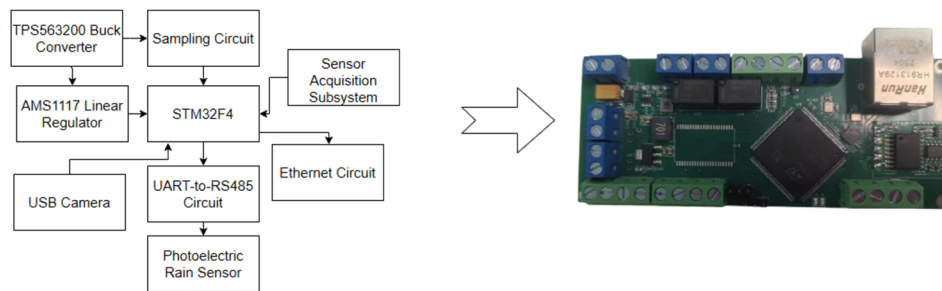


**Figure 3.** Physical Implementation of Multi-Network Convergence Circuit Module.

#### 3.3. Central Control System Circuit Design

The Central Control System circuit uses the STM32F407ZGT6 processor. As the core of the entire system, it needs to complete tasks including receiving data from the sensor acquisition submodule, transmitting data packets to the routing module, monitoring power supply voltage, managing the photoelectric rain sensor, and receiving data from USB cameras. Therefore, in addition to the essential minimum system circuit, interfaces such as USB, Ethernet, ADC, and serial communication ports must be reserved. Serial communication is used for connecting to subsystem circuits and the photoelectric rain sensor. The Ethernet interface transmits data to the routing module circuit, the ADC interface

monitors power supply voltage, and the main control chip has two USB ports: one connects to the internal bucket-status camera installed inside the tipping-bucket rain gauge to monitor its operational status, while the other connects to an external environmental camera that observes weather conditions and surroundings for subsequent manual data correction. Utilizing MJPEG-format cameras with typical frame sizes of 10-20 KB per frame. The Central Control System circuit block diagram and its physical implementation are shown in Figure 4.

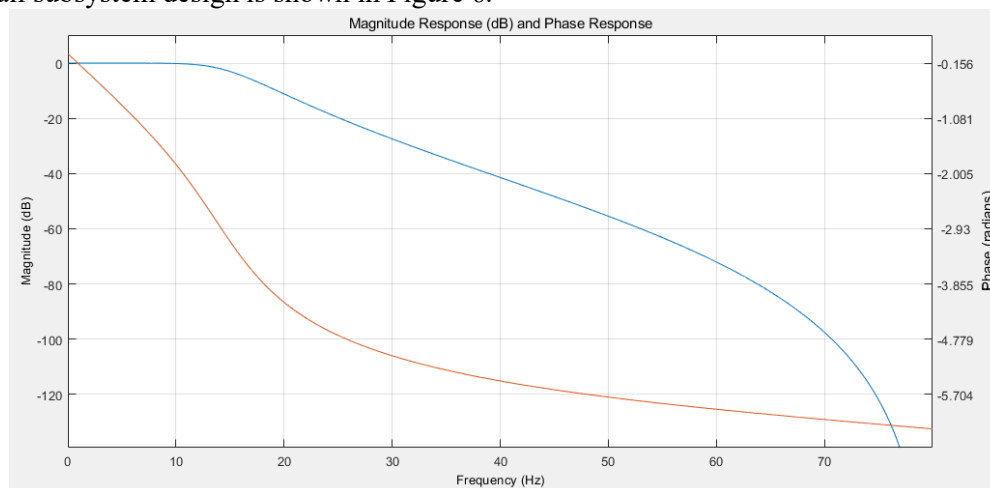


**Figure 4.** Central Control System circuit block diagram and its physical implementation.

## 4. System Software Design

### 4.1. Sensor Acquisition Module Software Design

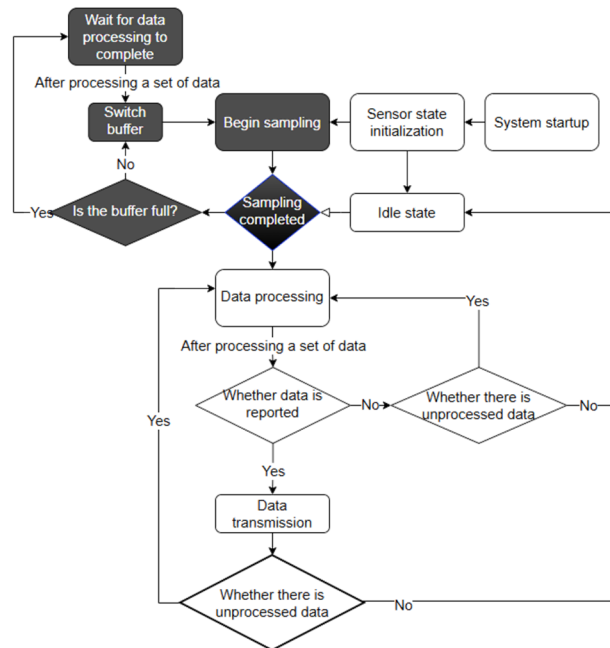
Multiple sensors monitoring the same tipping event allow mutual calibration of signals and collaborative validation of tipping action validity, improving the accuracy of obstruction detection. Subsequently, the MCU performs threshold determination, event judgment, and digital filtering on the data. Digital filtering addresses false tipping signals caused by mechanical vibration and impact during heavy rainfall events, thereby preventing data errors. A 4th-order Butterworth low-pass filter with a 15 Hz cutoff frequency attenuates characteristic 10-25 Hz shock vibrations (see Figure 5). Finally, the MCU packages the data according to a custom communication protocol and transmits it to the main control system via serial communication. The software flowchart for this sensor acquisition module in the overall subsystem design is shown in Figure 6.



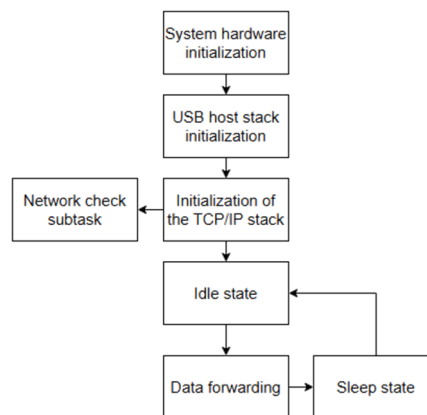
**Figure 5.** Bode plot of the low-pass filter

### 4.2. Multi-Network Convergence Routing Module Software Design

The Multi-Network Convergence Routing Module, serving as the communication bridge between the entire acquisition system and remote servers, is primarily responsible for critical tasks including data forwarding, link health status management, and network switching. The software flowchart for the Multi-Network Convergence Routing Module is shown in Figure 7.



**Figure 6.** Sensor Acquisition Module Software Flowchart



**Figure 7.** Multi-Network Convergence Routing Module Software Flowchart

#### 4.3. Central Control System Software Design

The Central Control System is the core control center of the entire acquisition system, operating on the STM32F407ZGT6 high-performance microprocessor, required to handle most functions of the acquisition system, including unpacking and fusing sensor data sent from sub-boards, receiving and parsing data from optical rain sensors, receiving and forwarding data from environmental cameras and tipping bucket status cameras to remote servers for further analysis, monitoring power supply voltage and I/O health status, and managing low-power modes of circuits.

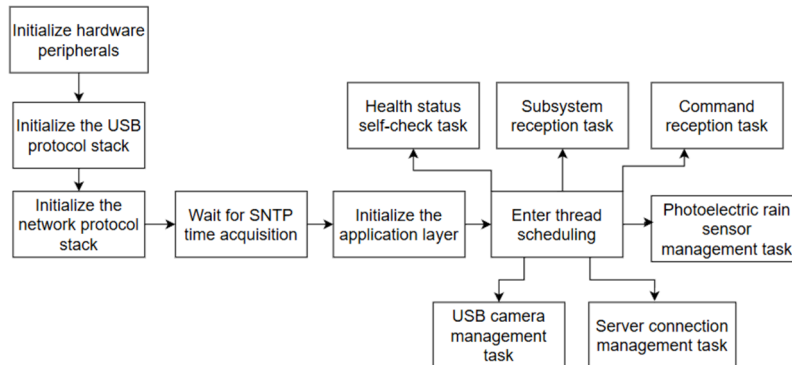
Additionally, the system implements a three-tier sensor fusion architecture: (1) Photoelectric rainfall sensor serves as the primary data source; (2) Infrared pairs compensate rainfall at 0.2mm per tip when valid obstruction ( $>15$  ms) is detected with photoelectric data timeout ( $\geq 5$  s); (3) Reed switch activates progressive arbitration: Dynamic threshold (see Equation 1) triggers low-confidence flagging and switches to backup mode after 3 consecutive exceedances.

$$\Delta = 0.2 + 0.05R \quad (1)$$

(R: real-time rainfall intensity)



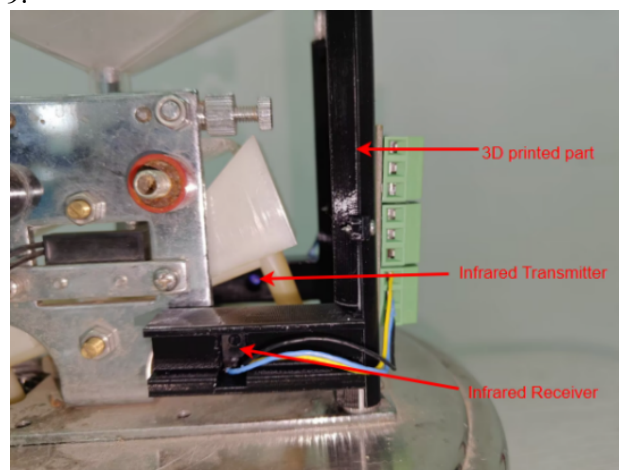
The logic of the entire Central Control System is complex; to make its logic clear, the software of the Central Control System incorporates an RTOS component, using modular software design. The software flowchart of the Central Control System is shown in Figure 8.



**Figure 8.** Central Control System Software Flowchart.

### 5.Full Machine Functional Debugging

Subsystem boards and sensor installation necessitate the use of 3D-printed support fixtures to adjust the position of infrared pairs, ensuring effective obstruction. The final adjusted position of the infrared pairs is shown in Figure 9.



**Figure 9.** Infrared Pair Position Diagram.

The UVC data packets from USB cameras are transmitted by splitting each frame into multiple packets. Since the Central Control System uses no external SRAM, packet loss testing is necessary. Connecting the device to a PC for transmission shows an extremely low packet loss rate. Quantitative testing was conducted by transmitting 1000 frames at 640×480 resolution using MJPEG compression over USB 2.0. The observed packet loss rate was less than 0.05%, and video frame rate remained stable at 25 fps.

Optical rain sensor testing: Testing is required to confirm normal acquisition of rainfall data and whether it can properly wake up other devices.

Self-check function testing: Adjusting input voltage triggers alert messages to remote servers when the voltage drops below the threshold.

Multi-network routing function testing: Removing antennas simulates poor network conditions. Observing debug outputs confirms that routers execute network switching operations.

Low-power function: The system employs power gating for power management. When entering low-power mode, measuring power supply voltages of all modules verifies successful entry into low-power mode. Power consumption dropped from 1.45 W (active) to 0.37 W (low-power mode). Peripheral module voltages confirmed shutdown of all non-critical systems, validating power gating success.

To validate field performance, the assembled rain gauge underwent 120 field trials during the 2024 flood season in Sichuan Province (May to September). Key error metrics are summarized in Table 1:

**Table 1.** Comparison of Key Error Metrics Between the Traditional Gauge and the Proposed Systems.

| Test Scenario        | Traditional Gauge Error | Proposed System Error |
|----------------------|-------------------------|-----------------------|
| Heavy Rain (>50mm/h) | 12.8%                   | 3.2%                  |
| Thunderstorm         | 23.1%                   | 5.7%                  |
| Strong EMI           | 17.4%                   | 4.1%                  |

## 6.Summary

This paper addresses the issues of significant data inaccuracies and weak anti-interference capabilities in traditional dual-tipping-bucket rain gauges during practical use, proposing and designing a multi-source fusion-based dual-tipping-bucket rainfall acquisition system. The system adopts a modular design approach, layering the entire system into acquisition subsystems, a Central Control System, and a Multi-Network Convergence Routing Module. Employing the high-performance STM32F4 microprocessor as its core unit, the Central Control System integrates multiple sensors, including infrared pairs, reed switches, optical rain sensors, and dual USB cameras, establishing a multi-source perception system for rainfall data.

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